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GitHub Link	https://github.com/120608akhila/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO3

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 1
Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0405**
SLOT : D

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25% **Duration: 90 minutes**
Assessment Tool Weightage: 25% **Marks: 25**

Code of Conduct:

*I **P.AKHILA(192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Akhila.p **Name of the student:**AKHILA POCHIMIREDDY

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS:02 **Total Marks: 20**

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)
A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

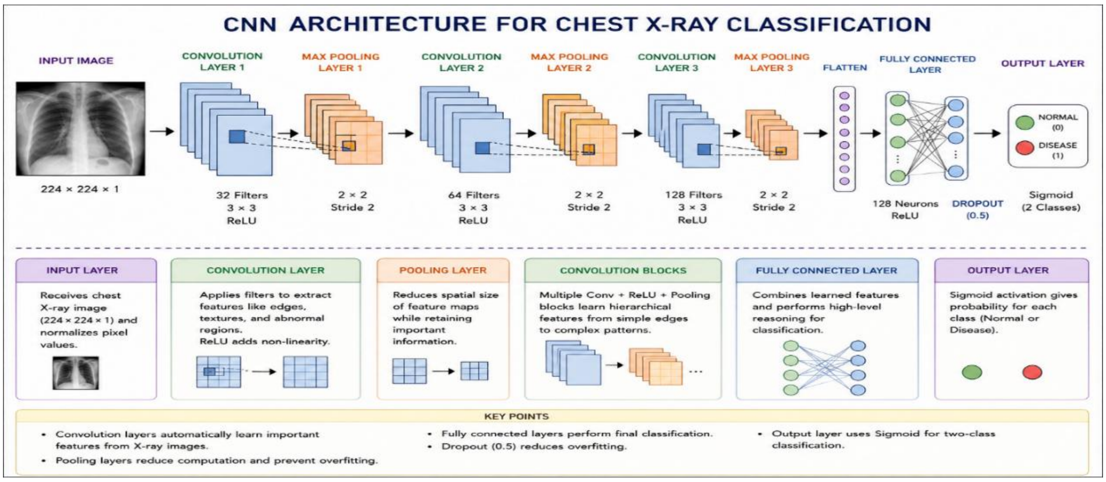
1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

1. Suitable CNN Architecture

A suitable CNN architecture for chest X-ray classification can be:

- **Input layer:** Accepts resized chest X-ray images, for example, 224 × 224 × 1.
- **Convolutional layers:** Extract important features such as edges, textures and abnormal regions.
- **ReLU activation:** Introduces non-linearity and helps the network learn complex patterns.
- **Pooling layers:** Reduce the spatial size of feature maps and computational cost.
- **Fully connected layer:** Combines the learned features for classification.
- **Dropout:** Reduces overfitting.
- **Output layer:** Uses **sigmoid** for two classes: Normal and Disease.

This architecture is suitable because CNNs automatically learn visual features from X-ray images without requiring manual feature extraction.



2. Motivation and Role of CNN Components

CNNs are suitable for medical images because they can automatically learn hierarchical visual features.

- **Convolutional layer:** Applies filters to the image and extracts useful features.
- **Filters:** Detect features such as edges, textures, shapes and abnormal structures.
- **Pooling layer:** Reduces image dimensions while retaining important information.
- **Fully connected layer:** Uses the extracted features to make the final classification.
- **Activation function:** ReLU helps the network learn non-linear patterns.

The early layers generally learn simple features, while deeper layers learn more complex medical patterns.

3. Parameter Sharing

In a fully connected network, every neuron can connect to every pixel, resulting in a very large number of parameters.

CNNs use **parameter sharing**, where the same filter weights are applied across different regions of an image.

For example, a **3 × 3 filter** has only 9 weights (plus bias), and the same weights are reused throughout the image.

Therefore, parameter sharing:

- Reduces the number of trainable parameters.
- Reduces memory requirements.
- Reduces computational cost.
- Allows the same feature to be detected at different locations.

Thus, CNNs are much more efficient for image processing than fully connected networks.

4. Regularization Techniques

Medical X-ray datasets may contain limited variations, so the model can overfit the training data. The following techniques can be used:

Dropout:
Randomly disables some neurons during training. This prevents the network from depending too much on particular neurons.

Data Augmentation:
Creates modified training images using operations such as rotation, shifting, scaling, cropping and brightness adjustment. This increases data diversity.

Batch Normalization:
Normalizes intermediate activations during training, making training more stable and helping generalization.

These methods help the CNN perform better on unseen X-ray images.

5.AlexNet vs ResNet — Suitable Choice

ResNet would be more appropriate for this medical image classification system.

Factor	AlexNet	ResNet
Architecture	Older CNN architecture	Modern deep CNN
Depth	Relatively shallow	Can be very deep
Feature learning	Good	More powerful
Training	Easier but limited	Skip connections improve training
Vanishing gradient	More susceptible	Skip connections reduce the problem
Performance	Good baseline	Generally better for complex images
Computational cost	Lower	Higher depending on version

Justification: Chest X-rays contain complex patterns, so deeper feature learning is useful. ResNet uses **residual/skip connections**, allowing deeper networks to be trained effectively and reducing vanishing-gradient problems.

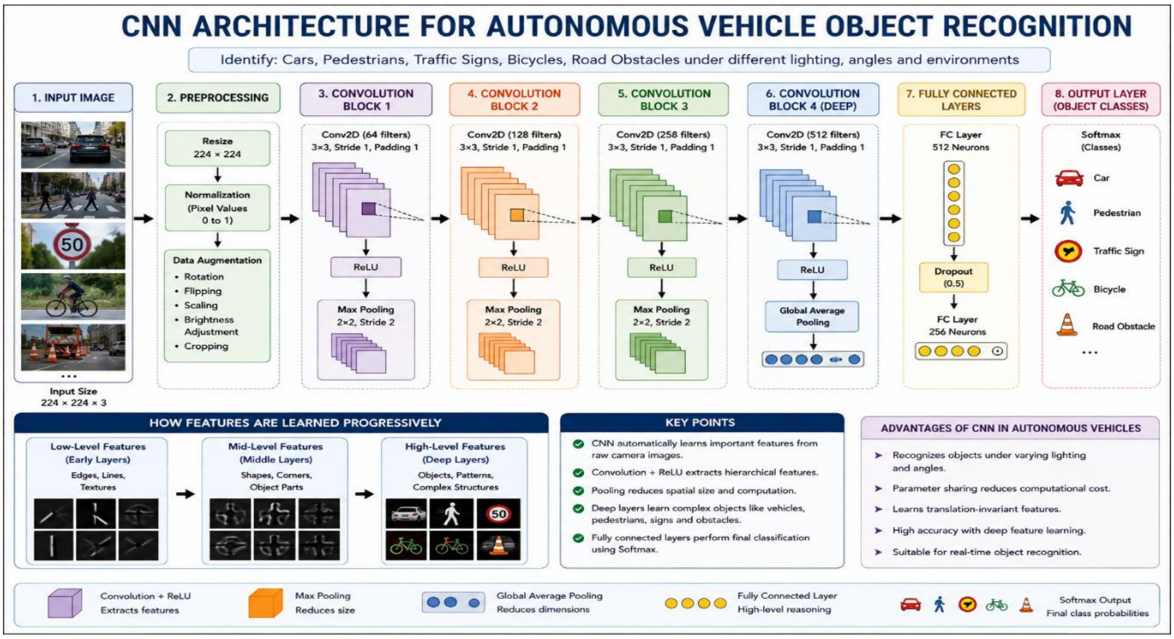
Therefore, **ResNet is the better choice when classification performance and feature learning are the main priorities**. A lighter ResNet variant can be selected if computational resources are limited.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)
An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.
Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

1. CNN Architecture for Autonomous Vehicles

A suitable architecture can be:



The system receives images from vehicle-mounted cameras and identifies:

- Cars
- Pedestrians
- Bicycles
- Traffic signs
- Road obstacles

The network should be trained using images captured under different lighting conditions, viewing angles, distances and road environments.

2. Purpose of CNN Layers

Input Layer:
Receives camera images and converts them into a suitable format for processing.

Convolution Layer:
Uses filters to extract important image features.

Activation Layer:
ReLU introduces non-linearity and helps the network learn complex patterns.

Pooling Layer:
Reduces the spatial dimensions of feature maps and decreases computation.

Fully Connected/Output Layer:
Uses the learned features to determine the object category or detection output.

Progressive Feature Learning

CNN filters learn features progressively:

Early layers → Edges and simple textures

Middle layers → Shapes and object parts

Deep layers → Complete objects and complex patterns

For example, the network may first detect edges, then wheels or human shapes, and finally recognize a **car or pedestrian**.

This hierarchical feature learning makes CNNs suitable for autonomous vehicle vision systems.

3. Importance of Parameter Sharing

Parameter sharing means that the same convolutional filter is applied at different locations of an image.

It provides two major benefits:

1. **Reduces computational complexity** because fewer unique parameters need to be learned.
2. **Improves translation-related feature detection** because the same feature can be detected regardless of where it appears in the image.

For example, a filter trained to detect a particular edge can detect that edge on the left, center or right side of a road image.

Therefore, parameter sharing is important for recognizing objects appearing at different positions.

4. Overfitting and Regularization

The autonomous vehicle dataset may contain many images, but the model can still overfit to particular roads, weather conditions or camera viewpoints.

Suitable techniques include:

- **Data augmentation:** Rotation, scaling, cropping, brightness changes and other realistic transformations.
- **Dropout:** Randomly removes neurons during training.
- **Batch normalization:** Stabilizes network training.
- **Early stopping:** Stops training when validation performance stops improving.
- **Weight regularization:** Penalizes excessively large model weights.

These methods improve the model's ability to generalize to new road environments.

5.AlexNet vs ResNet

Feature	AlexNet	ResNet
Depth	Relatively shallow	Much deeper
Architecture	Traditional CNN	Residual CNN
Parameter efficiency	Less efficient for deeper models	Better deep-network design
Training difficulty	Simpler	Skip connections make deep training easier
Vanishing gradient	More affected	Reduced using skip connections
Feature learning	Good	Stronger hierarchical feature learning
Modern applications	Limited compared with newer networks	Widely suitable for complex vision tasks

AlexNet introduced important CNN techniques and works well as a basic image classifier.

ResNet uses shortcut connections:

Input → Convolution layers → Add original input → Output

These residual connections help information and gradients flow through deep networks, making it possible to train deeper models effectively.

6. Recommended Architecture

ResNet is the most suitable choice for the autonomous vehicle application.

Reasons:

- **Higher accuracy:** Better feature learning for complex road scenes.
- **Deep feature extraction:** Can learn detailed representations of vehicles, pedestrians and signs.
- **Reduced vanishing-gradient problem:** Residual connections make deep training easier.
- **Robust recognition:** Better suited to different viewing angles and environments.
- **Real-time possibility:** A lightweight version such as ResNet-18 can provide a better balance between accuracy and computational cost.

Although ResNet can require more computation than AlexNet, an appropriately selected ResNet variant provides a strong balance between **accuracy, feature-learning capability and real-time requirements**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand